

Lab 10

MOSFET Common Source Amplifier

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10.1 Objective

To investigate the transfer characteristics of MOSFET and characteristics of MOSFET common source amplifier

10.2 MOSFET Transfer Characteristics

Consider the circuit shown in Figure 1. Here, an n-channel MOSFET is in common source configuration. If it is viewed as two-port network, the source terminal is common to input and output sides. We have obtained the output characteristic curve (I_d as a function of V_{ds}) and the input characteristic curve (I_d as a function of V_{gs}) in previous labs. The effect of change in drain current due to change in gate-to-source voltage results in change in drain-to-source voltage when a resistor R_D is used as shown in the Figure 1.

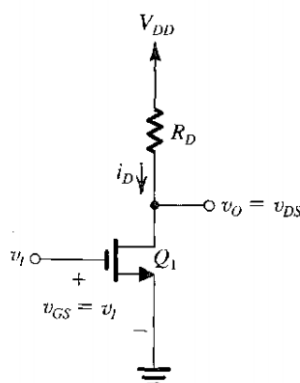


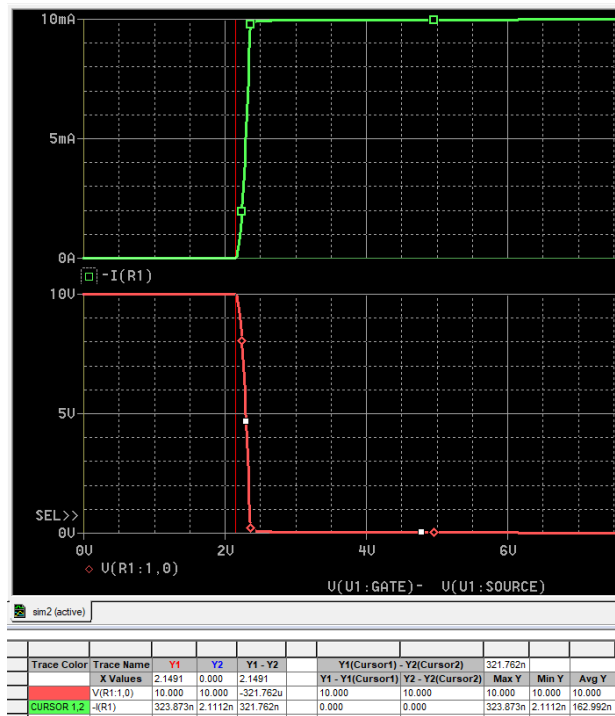
Figure 1: MOSFET in common source configuration

We will use this circuit to determine the relation between its output voltage V_o and its input voltage V_i . These characteristics are called transfer characteristics of MOSFET. The output voltage can be given as: $V_o = V_{DS} = V_{DD} - I_D R_D$

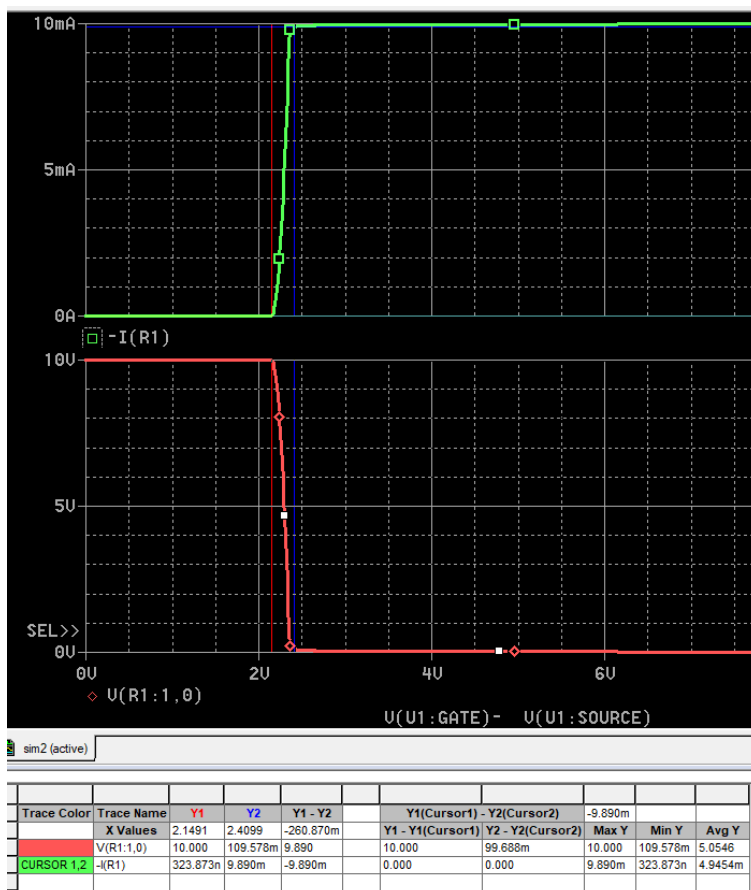
Task 1: To obtain transfer characteristic curve of n-channel MOSFET

1. Use 2N7000 n-channel MOSFET and set $V_{DD}=10V$. Connect R_D equal to $1k\Omega$ to implement the circuit of Figure 1 in PSpice. Connect a separate DC source at input side (gate terminal). Vary this input voltage from 0 to 10V through sweep analysis.
2. Obtain transfer characteristic curve i.e., plot of V_{DS} as a function of V_{GS} . Identify and label the regions of operation on the curve and add capture of the labelled plot. Note that the threshold voltage of 2N7000 is 2.14V as determined in previous labs.

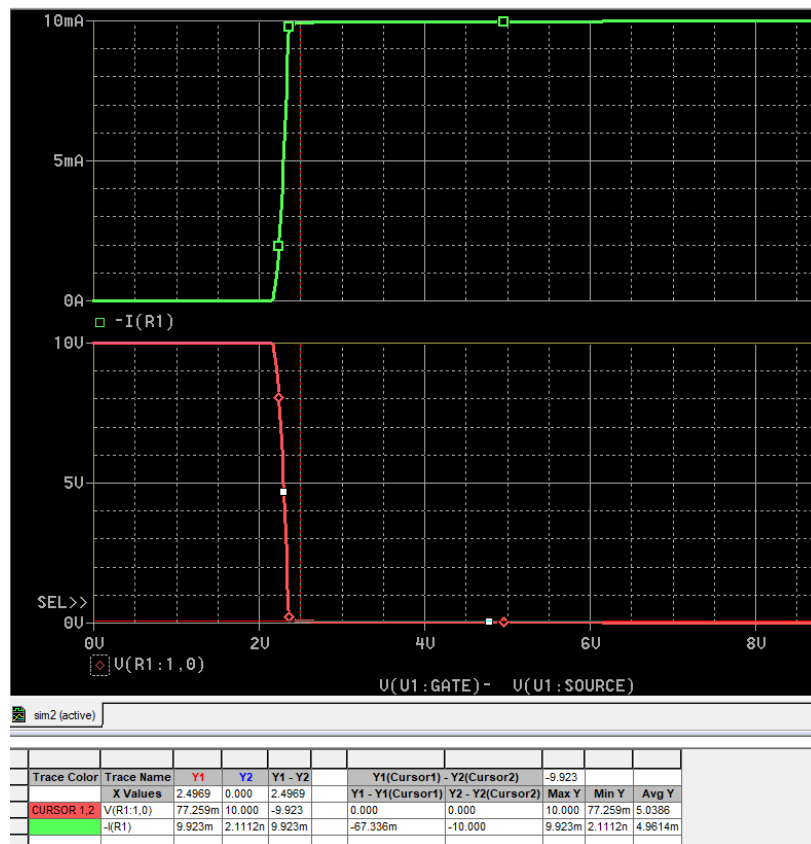
CUT OFF:



SATURATION:



TRIODE REGION:



10.3 Introduction to MOSFET Amplifier

From the transfer characteristic curve of MOSFET, it can be seen that in saturation region, there is a large variation in drain-to-source voltage for small change in gate-to-source voltage. In this region, MOSFET can be used as an amplifier where small AC signal is superimposed on input gate terminal and output is taken across drain terminal. The basis for this important MOSFET application is that when operated in the saturation region, the MOSFET acts as a voltage-controlled current source: changes in the gate-to-source voltage gives rise to changes in the drain current and using a resistor R_D , amplified output voltage V_O is obtained.

This effect of variation in gate-to-source voltage can be related to the change in Q-point in BJT output characteristic curves along load line when small AC signal is superimposed on input base terminal. To operate MOSFET as an amplifier, it is first biased in saturation region such that the DC operating point is at center to allow maximum swing in both directions. This is represented in Figure 2.

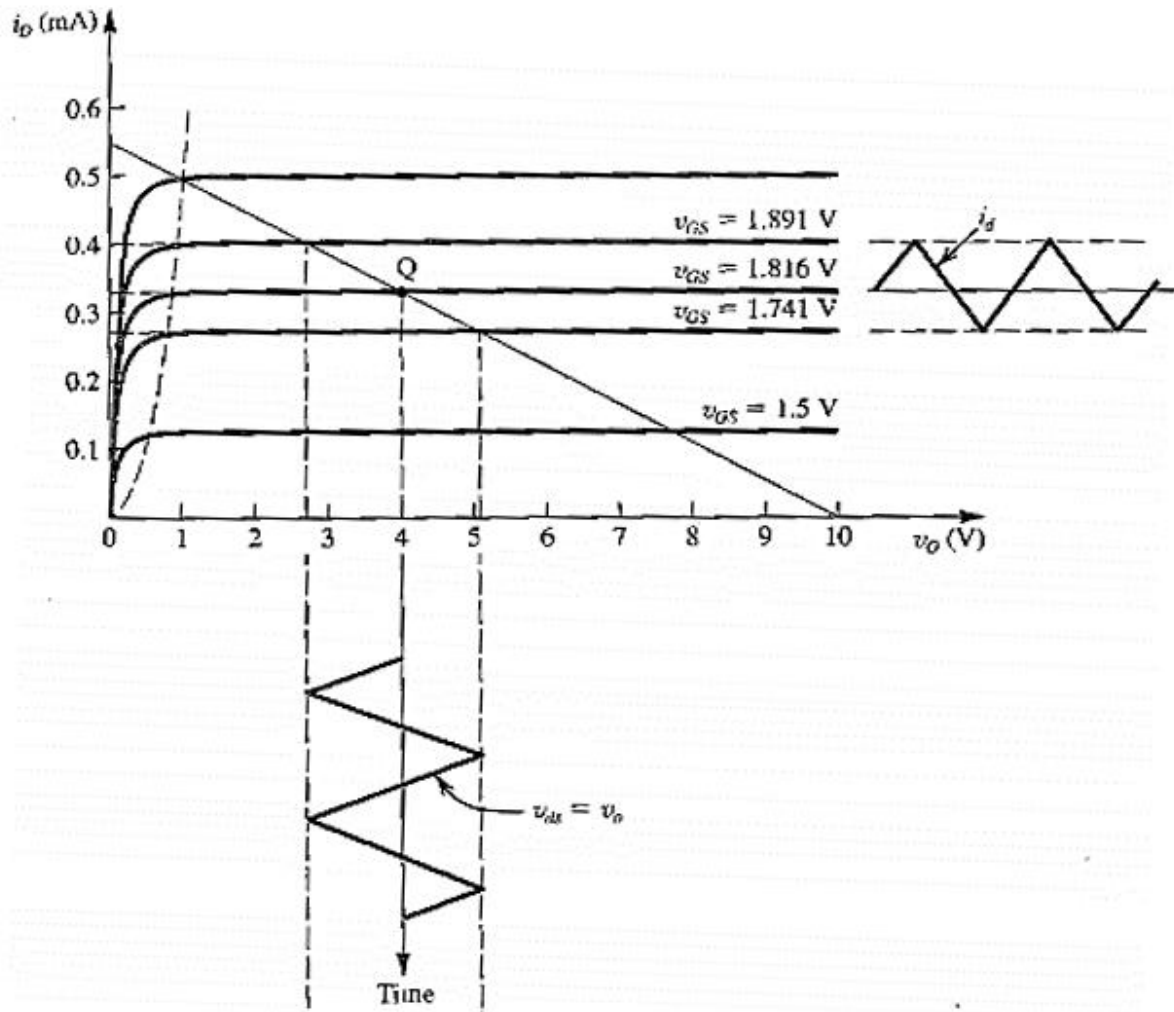


Figure 2: Amplification by MOSFET in saturation region

Task 2: To bias MOSFET in saturation region and obtain DC operating point.

1. Implement the circuit shown in Figure 3. In this circuit, voltage-divider biasing configuration is used for implementation of CS MOSFET amplifier. For now, connecting capacitors is not required as AC input is not applied.
2. Perform bias point analysis to find the DC operating point of MOSFET and collect relevant data to complete Table 1. Verify the mode of operation of MOSFET.

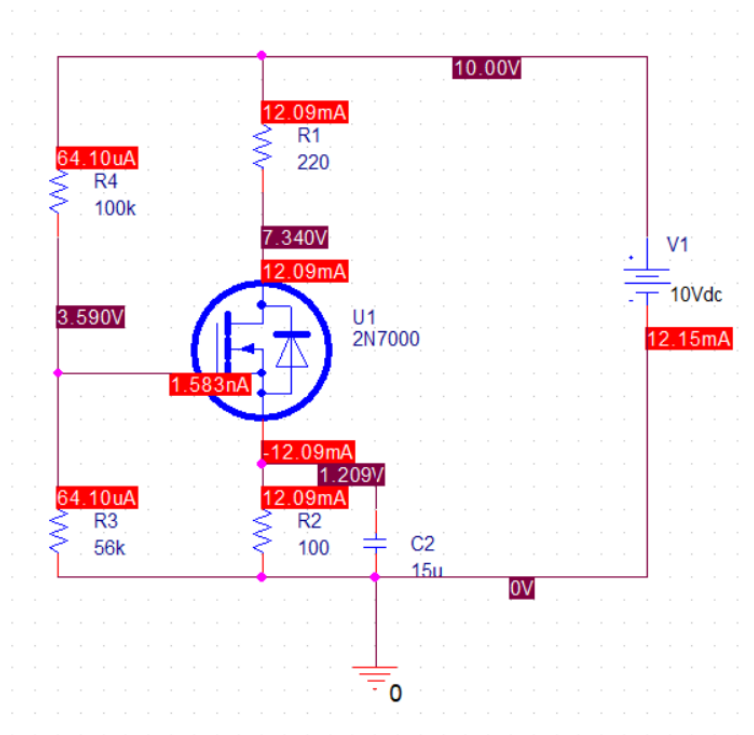


Table 1: DC analysis of CS MOSFET amplifier circuit

Parameters	Measured Value
Drain Current	12.09mA
Drain Voltage	7.34V
Gate Voltage	3.59V
Source Voltage	1.209V
Drain-to-Source Voltage	6.131V
Gate-to-Source Voltage	2.381
Mode of Operation	Saturation Region

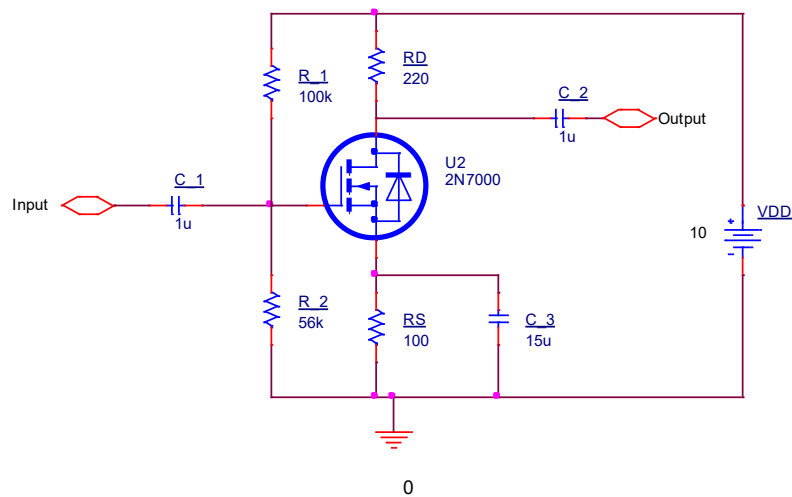


Figure 3: Common source MOSFET amplifier circuit

2N7000 MOSFET Pinout

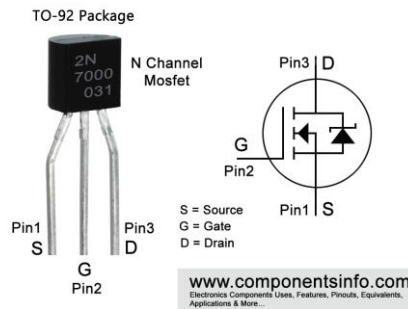


Figure 4: 2N7000 Terminals

10.4 AC Analysis of Common Source MOSFET Amplifier

The MOSFET behaves as a voltage-controlled current source in saturation region. It accepts a signal V_{gs} between gate and source and provides a current $g_m V_{gs}$ at the drain terminal. g_m is the transconductance of MOSFET as determined in Lab 08 using I_d Vs V_{gs} curve. The input resistance of this controlled source is very high—ideally, infinite. The output resistance r_o that is, the resistance looking into the drain—also is high, and is infinite for ideal MOSFETs in saturation region. This is highlighted in Figure 4 by blue box. `

For AC analysis, MOSFET in figure 3 is replaced by small signal model of MOSFET. R_G is the equivalent resistance of $(R1 || R2)$ used for biasing. R_L will be included in circuit only when amplifier is loaded.

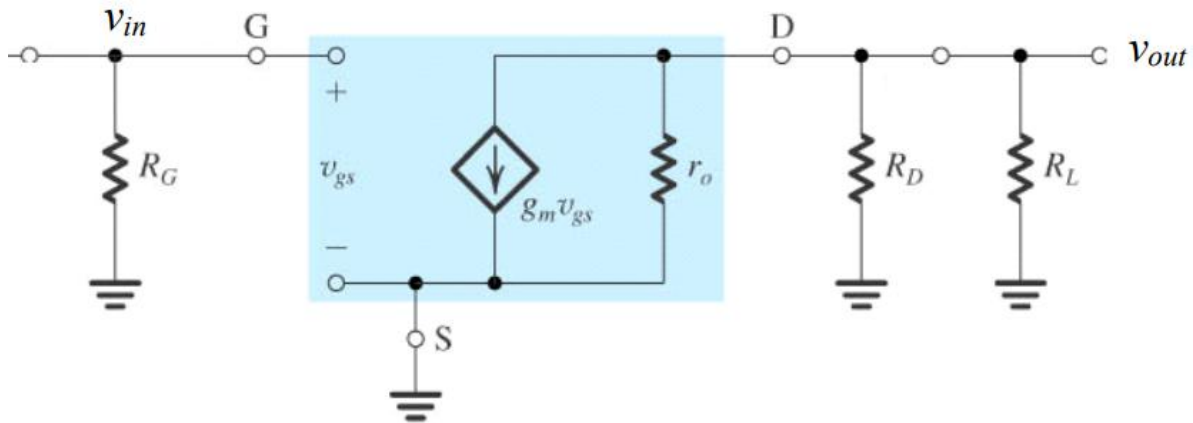


Figure 5: Small signal model of the circuit shown in Figure 3

Characteristic parameters of CS amplifier can be given as:

- AC Voltage Gain A_v is defined as $A_v = \frac{V_o}{V_i}$

Here, V_o and V_i are output and input voltages which can be measured as rms, peak or peak-to-peak. From model, the AC voltage gain of a CS amplifier (under no-load) can be calculated using:

$$A_v = -g_m(R_D \parallel r_o) = -g_m R_D$$

Where, transistor trans-conductance g_m can be calculated as

$$g_m = \frac{2I_D}{V_{GS} - V_T}$$

- Input Resistance, R_i , is that of the amplifier as seen by the input signal and mathematically it can be given as

$$R_i = R_G$$

- AC Output Resistance, R_o , for CS amplifier can be approximated to

$$R_o = R_D \parallel r_o = R_D$$

Task 3: To calculate CS MOSFET amplifier characteristic parameters

- Using the values obtained by DC analysis in Table 1, estimate the characteristic parameters of MOSFET amplifier and fill in Table 2.

Calculation

For g_m :

$$g_m = \frac{2 \times 12.09 \text{ mA}}{0.281} \approx 0.086 \text{ S}$$

FOR A_v :

$$A_v = -g_m \times R_D = -(0.0860 \times 220) = -189.2$$

for input resistance:

$$R_i = \frac{R_3 \cdot R_4}{R_3 + R_4}$$

$$R_i = \frac{(56k)(100k)}{56k + 100k}$$

$$R_i = \frac{5600000}{156000} = 35.9 \text{ k}\Omega$$

Table 2: AC analysis of CS MOSFET amplifier circuit

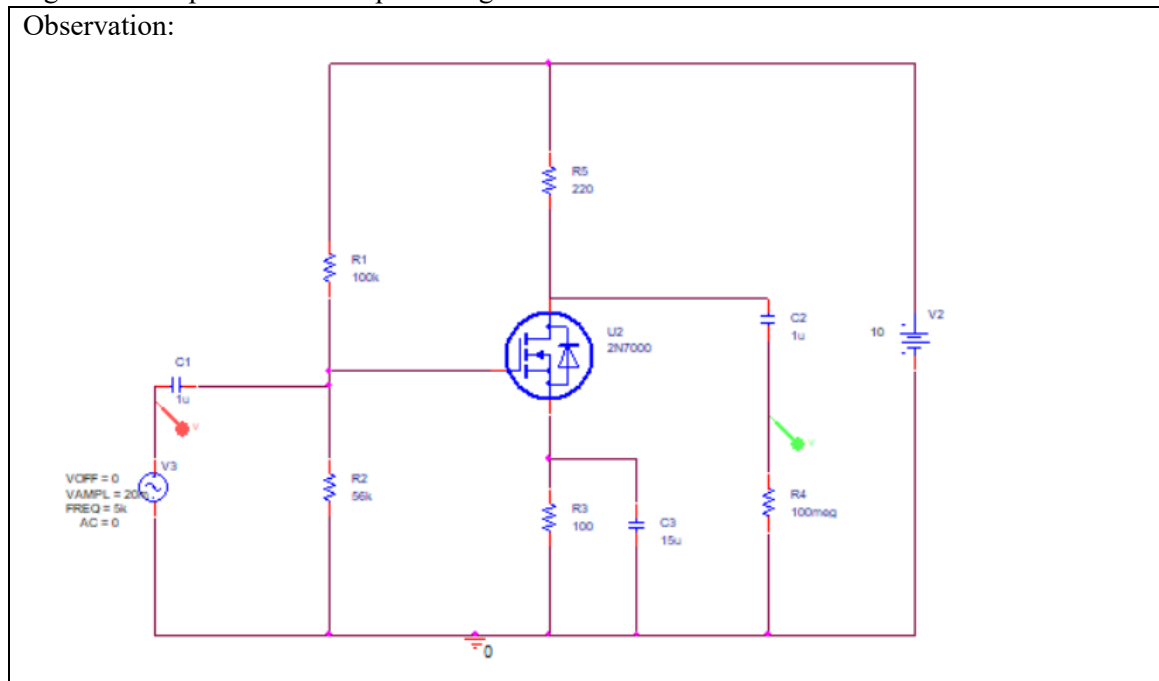
Characteristic Parameters	Estimated Values
Trans-conductance: g_m	0.0860
Voltage Gain: A_v	-18.92
Input Resistance: R_i	35.9k
Output Resistance: R_o	220

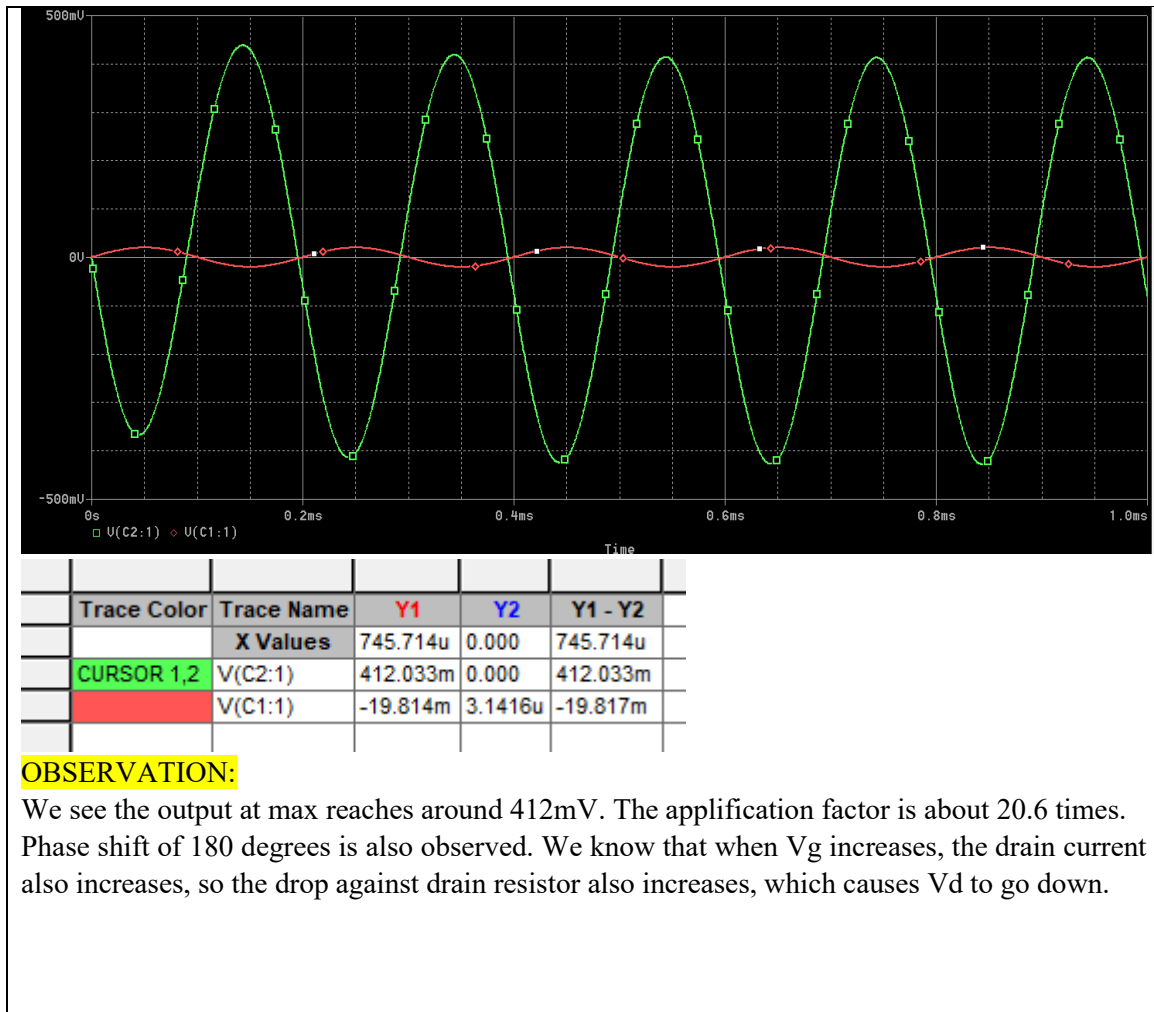
Task 4: To implement CS MOSFET amplifier circuit and experimentally determine its characteristic parameters

• Amplifier Voltage Gain A_v

1. Modify the implemented circuit to apply AC input of 20mV rms at 5 kHz frequency to the amplifier.
2. Obtain input and output voltage waveforms. Add capture of the obtain plots and comment on magnitude and phase of the output voltage.

Observation:





3. Compute voltage gain (unloaded) and fill in Table 3. Associate sign to represent phase shift.

• Amplifier Input Resistance R_i

To determine amplifier input resistance, we can consider the two-port network model of amplifier as shown in Figure 5. Here, the input impedance Z_i or R_i is the resistance seen from the input port of amplifier which is the ratio of input voltage and input current.

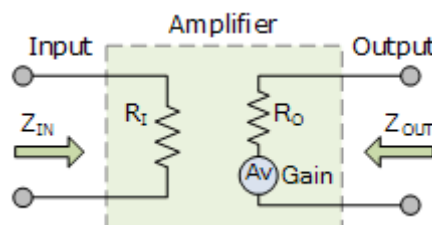


Figure 6: Representation of amplifier input and output resistance

When a source is connected at this input port, the network can be represented by Figure 6. This is used to determine the input resistance using voltage divider rule if source voltage V_s and source resistance R_s are known. Since we have ideal AC input voltage source, we can connect an external resistance to experimentally determine this unknown input resistance of amplifier.

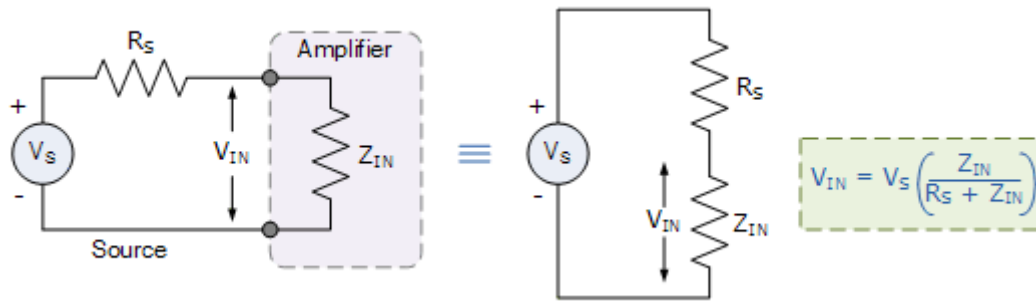
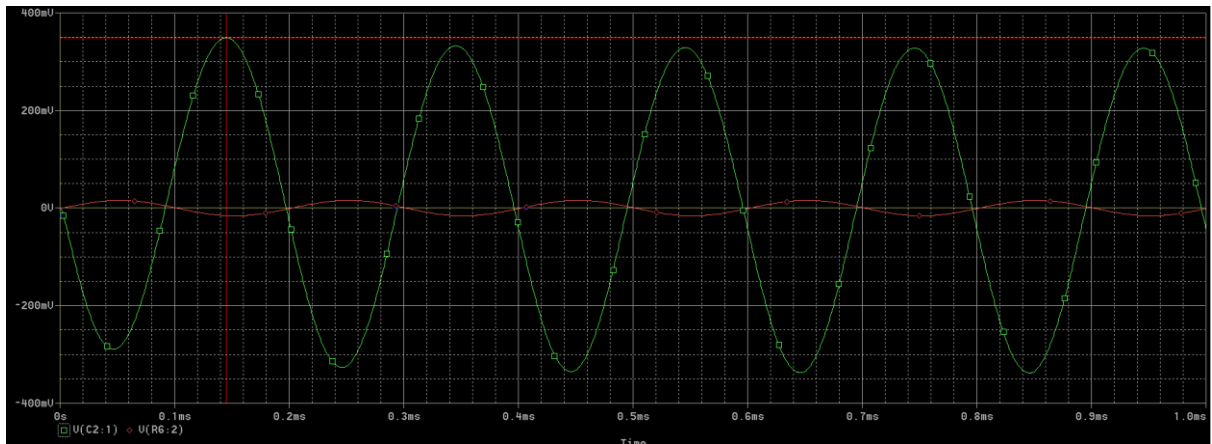
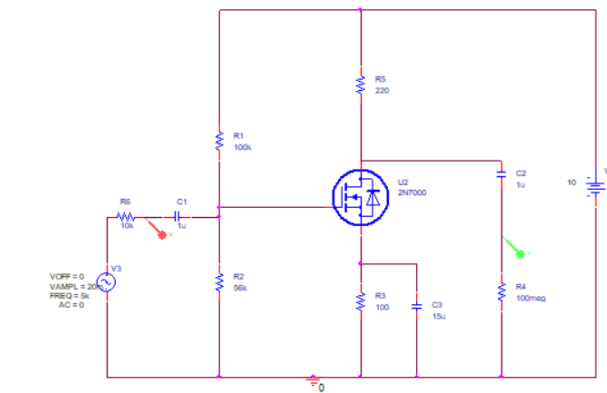


Figure 7: Circuit for experimental determination of input impedance of amplifier

1. Connect a measurement resistance ($R_s=10k\Omega$) at input side of amplifier after AC source with V_{sig} 20mV rms (same as previous part). Measure the voltage V_i input to the amplifier after this resistor.



2. Note down the measurements in Table 3 and calculate the input resistance of amplifier.

• Amplifier Output Resistance R_o

When amplifier is loaded (load resistor R_L is connected at the output port), the output side network of amplifier can be represented by Figure 7. When output voltage and load resistor are known, the amplifier output impedance can be determined using voltage divider rule.

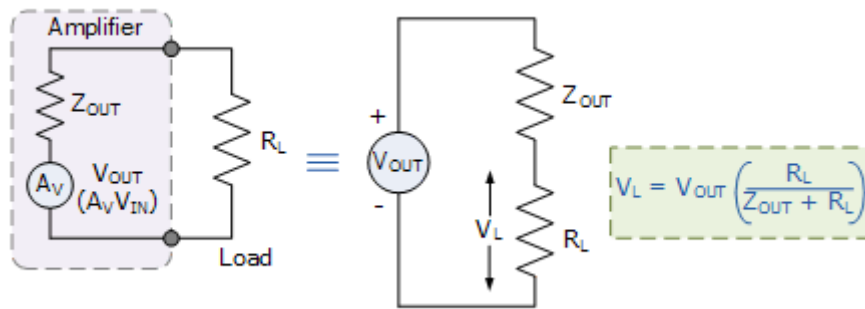


Figure 8: Circuit for experimental determination of amplifier output impedance

1. Remove the input measurement resistance once again and directly apply the AC voltage from source to amplifier input terminal.
2. To determine the amplifier output impedance, connect a load resistor of $1\text{k}\Omega$ at the output port of amplifier.
3. Measure this loaded output voltage and note down in Table 3. Using the collected data, compute the output resistance R_o .

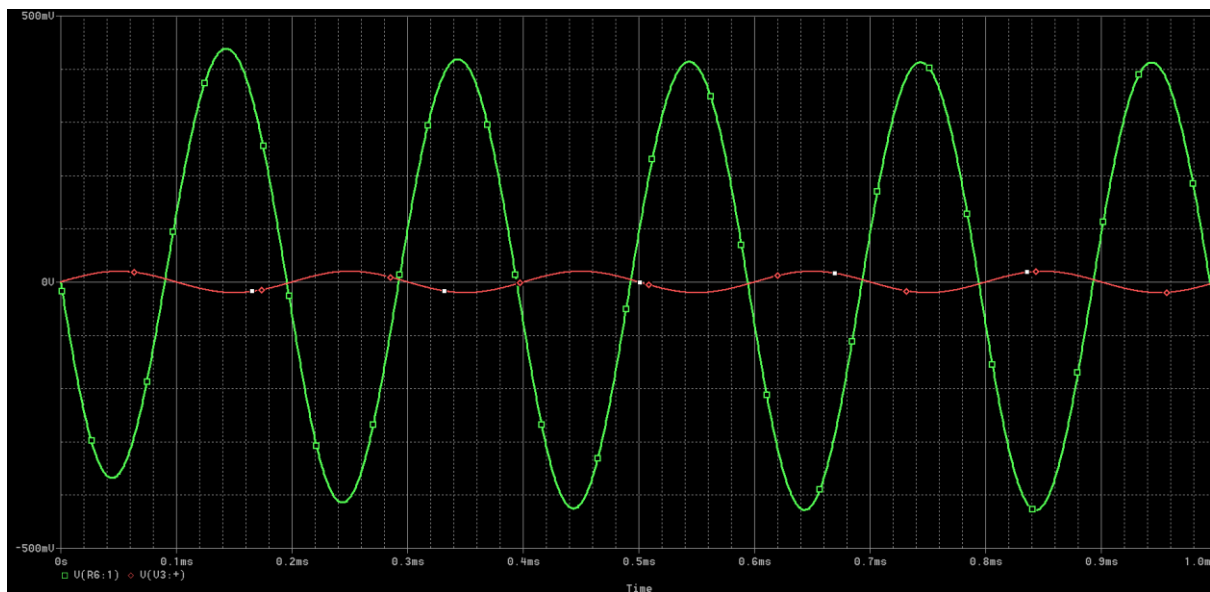




Table 3: Data table for experimental determination of CS amplifier characteristic parameters

Characteristic Parameter	Values	Required Measurements	Measured Values
Amplifier Voltage Gain: A_v (Unloaded)	20.6	Input Voltage: V_i	20mV
		Output Voltage (Unloaded): V_o	412mV
Amplifier Input Resistance: R_i	37.62k	Measurement Resistor: R_s	10k
		Source Voltage: V_{sig}	20mV
		Input Voltage: V_i	15.86mV
Amplifier Output Resistance: R_o	216	Output Voltage Unloaded: V_o	412mV
		Load Resistor: R_L	1k
		Output Voltage Load: V_L	338mV

4. Compare the experimentally determined and calculated values of gain, input resistance and output resistance of CS amplifier.

Voltage Gain: 18.92 vs 20.6 = 9% difference

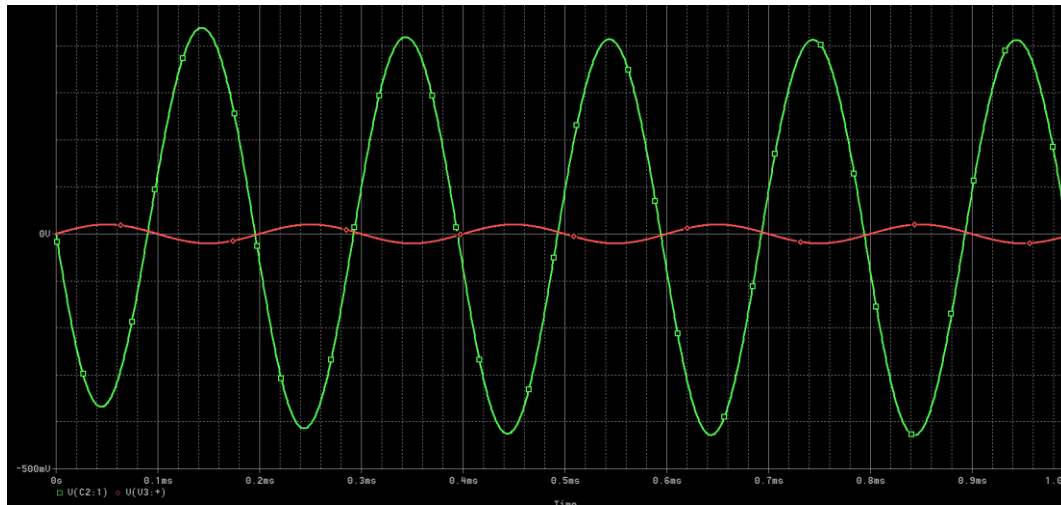
Input Resistance: 35.9k Ω vs 37.62k Ω = 4.8% difference

Output Resistance: 220 Ω vs 216 Ω = 1.9.% difference

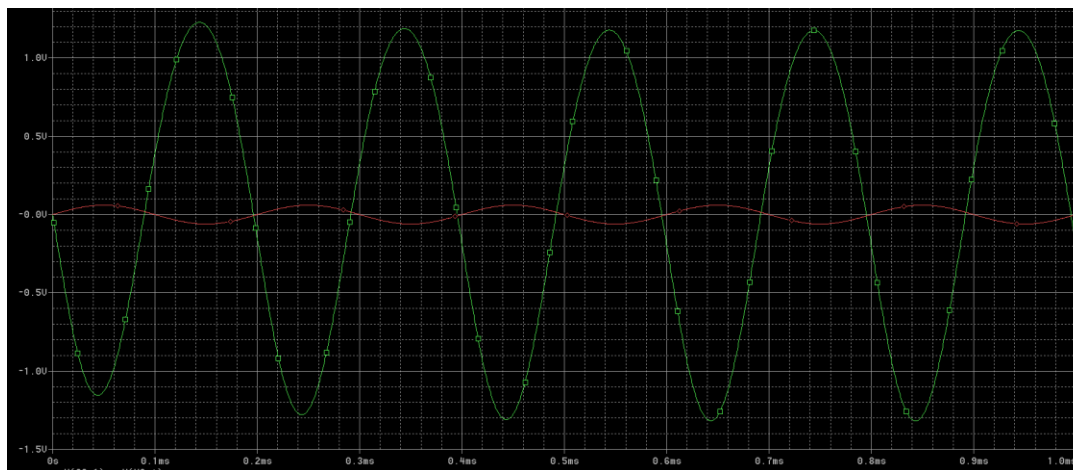
5. For the no load case, vary the input signal from 20 mV to 150 mV and observe the changes.

The amplifier operates linearly for input signals below 80-100 mV. When the input amplitude exceeds the linear range, the MOSFET transitions from the saturation region into either the triode region (during positive input peaks) or the cutoff region (during negative input peaks), resulting in clipped or distorted output. This demonstrates that proper DC biasing is critical for maximizing the amplifier's linear operating range.

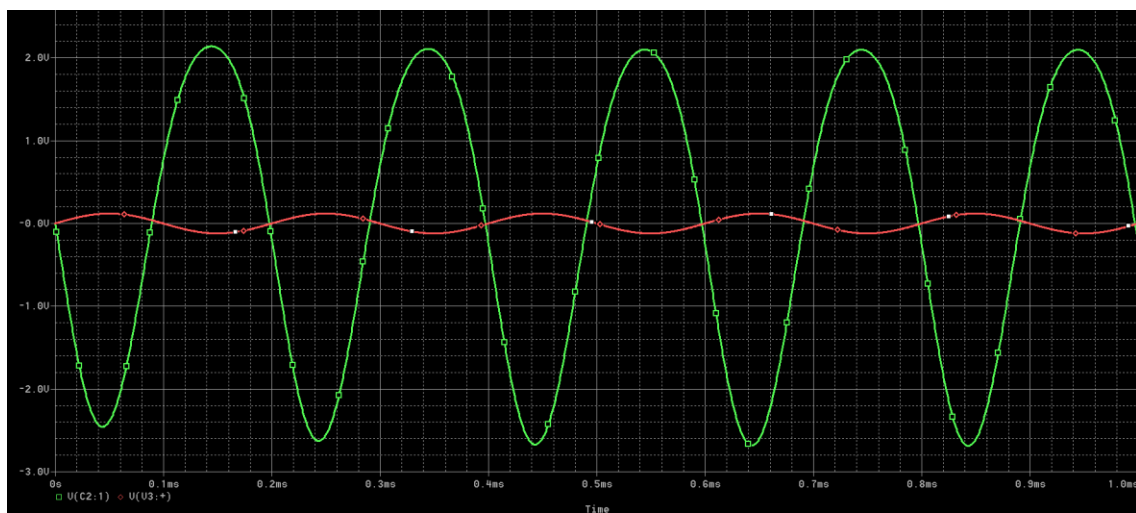
20m input:



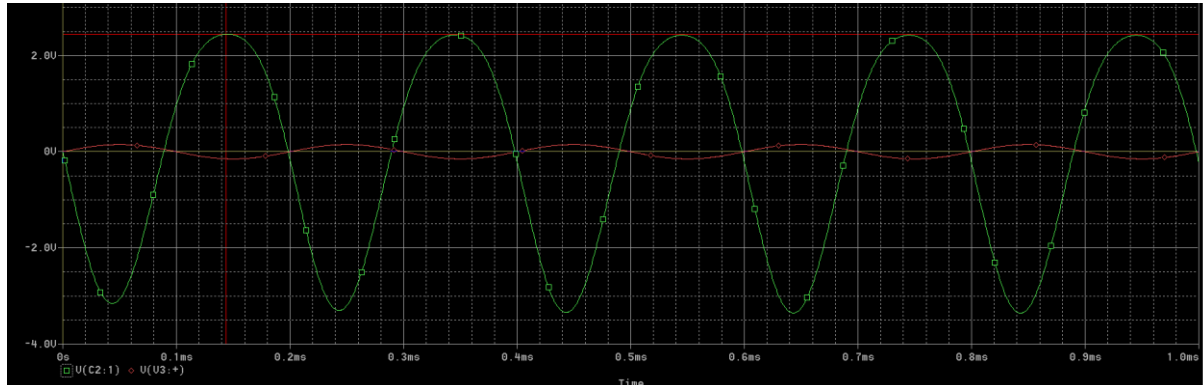
60mV input:



120mV input:



150mV input:



Task 5: To examine the effect of loading on CS amplifier voltage gain

- Using the measured loaded amplifier output voltage V_L (noted in Table 3 when $1k\Omega$ load resistor is connected), determine loaded gain of amplifier A_{VL} i.e., the ratio of output voltage and input voltage.
- Write mathematical expression to determine amplifier loaded gain in terms of circuit parameters from the small signal model of Figure 5 and use this expression to compute loaded voltage gain. Compare your experimental and mathematical values of loaded voltage gain.

$$A_{VL} = V_L / V_i = 339mV / 20mV = 16.95$$

Table 4: Determination of amplifier loaded voltage gain

Loaded Voltage Gain (Experimental)	-16.95
Loaded Voltage Gain (Expression)	$A_{VL} = -g_m \times [(R_D \times R_L) / (R_D + R_L)]$
Loaded Voltage Gain (Calculated)	-15.51

-ve sign means 180 degree phase shift.

- Comment about the effect of loading on amplifier gain.

Loading reduces the CS amplifier voltage gain by approximately 18% (from 20.6 to 16.95). This occurs because the load resistor R_L appears in parallel with the drain resistor R_D , reducing the effective output resistance. The loaded gain is 82% of the unloaded gain, demonstrating that proper impedance matching is essential for maintaining amplifier performance in practical applications



Post-Lab

Task 6:

By now, we have learned to use both BJT and MOSFET as an amplifier, what is the difference between MOSFET and BJT Amplifier? Which one is better and why?

The main difference is how they're controlled: BJTs need actual current flowing into the base to work, while MOSFETs just need voltage at the gate with basically zero current draw.

Key Differences:

- **Input Impedance:** MOSFETs have high input resistance, so they barely load down your signal source. BJTs are way lower than MOSFETs.
- **Power Consumption:** MOSFETs are very efficient since the gate doesn't pull any current. BJTs constantly need base current, which wastes some power.
- **Gain:** BJTs usually give you better voltage gain because they have higher transconductance.
- **Speed:** MOSFETs are faster at switching, which is why they dominate in digital circuits and power electronics.

Which is Better?

It completely depends on what you're doing. MOSFETs are great when you need low power consumption, high input impedance, or fast switching i.e. digital processes. BJTs are better when you need really high gain or precise linear amplification. Each one has its own use, there's no clear basis upon which one could be said better over the other.



Assessment Rubric

Lab 10

MOSFET Common Source Amplifier

Name:	Student ID:
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Points Distribution

Task No.	LR1/2 Simulation/Hardw are Performance	LR 4 Data Collection	LR 5 Plots	LR 6 Calculation	LR 10 Analysis
Task 1	/4	-	/8	-	-
Task 2	/8	/8	-	-	-
Task 3	-	-	-	/8	-
Task 4	/12	/8	/8	/4	/8
Task 5	-	-	-	/6	/6
(Post Lab) Task 6					/8
	/24	/16	/16	/18	/22
CLO Mapped	CLO 3	CLO 1	CLO 3	CLO 1	CLO 1

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	/4	CLO 4

CLO – (LDL)	Total Points	Points Obtained
CLO1 – Psy3	56	
CLO3 – Psy4	40	
CLO4 – Aff-3	04	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.